

A negative answer to a question of Carenini on almost independent sets in regular graphs*

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Abstract

For a d -regular graph G on n vertices and $\gamma \geq 0$, let $i_\gamma(G)$ be the number of vertex sets that span at most γdn edges. Carenini asked whether, when $2d \mid n$, the disjoint union of $n/(2d)$ copies of $K_{d,d}$ maximises i_γ among d -regular graphs on n vertices for every $\gamma \geq 0$. For $\gamma = 0$ this is the theorem of Kahn and Zhao on independent sets. We show that the answer is no. The smallest counterexample (smallest n , and smallest γ for that n) is $(n, d, \gamma) = (6, 3, 1/18)$: the triangular prism has 28 vertex sets spanning at most one edge, and $K_{3,3}$ has 24. More generally, for every pair (n, d) with $n > 0$ and $2d \mid n$, except the trivial cases $d = 1$ and $(n, d) = (4, 2)$ in which only one d -regular graph exists up to isomorphism, the answer is negative for some $\gamma < 1/2$; for $d \geq 3$ the union of copies of $K_{d,d}$ is then even a minimiser, and every d -regular graph containing a triangle beats it. For small γ we give two families: a switched $K_{d,d}$ beats $K_{d,d}$ on $2d$ vertices for every $d \geq 3$, and a connected bipartite d -regular graph beats two copies of $K_{d,d}$ on $4d$ vertices for every $d \geq 2$, so the answer is no even among bipartite graphs. For every fixed $\gamma \in (1/8, 1/2)$ and every $d \geq 2$, a bipartite competitor wins for all sufficiently large n . On the other hand, the union of copies of $K_{d,d}$ is optimal at the exponential scale for every γ ; the upper bound here is Carenini's biclique reduction, which rests on a theorem of Sah, Sawhney, Stoner and Zhao, which we also prove in the case used here. The exact question remains open for fixed $\gamma \in (0, 1/8]$ as $n \rightarrow \infty$; for $\gamma = 0$ and for every fixed $\gamma > 1/8$, the weaker statement in which the union of copies of $K_{d,d}$ is optimal up to a factor $1 + o(1)$ holds. Every result of the paper is formally verified in Lean 4. The counterexamples and proofs were found by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

Background. For a graph G and a set $A \subseteq V(G)$ let $e_G(A)$ be the number of edges of G with both ends in A . Following Carenini [2], for a d -regular graph G on n vertices and a real number $\gamma \geq 0$ put

$$i_\gamma(G) = |\{A \subseteq V(G) : e_G(A) \leq \gamma dn\}|,$$

so that $i_0(G)$ is the number of independent sets of G . Alon [1] proved that a d -regular graph on n vertices has at most $2^{(1/2+o(1))n}$ independent sets, where $o(1) \rightarrow 0$ as $d \rightarrow \infty$. Kahn [8] proved the sharp bound $i_0(G) \leq (2^{d+1} - 1)^{n/(2d)}$ for bipartite d -regular graphs, and Zhao [14]

* *Version 2.* Every result of this version is formally verified in Lean 4 (Section 9). Two claims of version 1 for which we have no formal proof have been deleted: the normal approximation at $\gamma = 1/8$ in Remark 10, with the corresponding entry of reading R6, and a count of labelled graphs in Section 8. A few other sentences have been reworded or removed; the theorems are unchanged. The main changes are listed at the end of Section 9.

extended it to all d -regular graphs. Since $K_{d,d}$ has $2^{d+1} - 1$ independent sets, this says that when $2d \mid n$, the disjoint union $mK_{d,d}$ of $m = n/(2d)$ copies of $K_{d,d}$ has the largest number of independent sets among d -regular graphs on n vertices.

In work on tolerant testing of independent sets, Seth [12] bounded the number of sets $J \subseteq V(G)$ whose edge density $e_G(J)/\binom{|J|}{2}$ is at most $d/(kn)$, and asked for the correct order of this bound and whether the disjoint union of $n/(2d)$ copies of $K_{d,d}$ is extremal. Carenini [2] proved a robust form of the Kahn–Zhao theorem for the count i_γ : there is an absolute constant C such that

$$i_\gamma(G) \leq 2^{n/2} \exp\{C\gamma \log(e/\gamma) n + Cn/d\}$$

for every d -regular graph G on n vertices with $d \geq 2$ and every $0 < \gamma \leq 1/2$ [2, Theorem 1.1], and both error terms are needed [2, Proposition 1.2]. As Carenini notes, Seth’s density condition implies $e_G(J) \leq dn/(2k)$, so this answers the quantitative part of Seth’s question. The key step [2, Proposition 2.1] is a biclique reduction based on a theorem of Sah, Sawhney, Stoner and Zhao [11, Corollary 1.15] (Corollary 1.14 in version 2 of the arXiv preprint, the numbering used in [2]): in the case used there, for $0 < q \leq 1$, the sum of $q^{e_G(A)}$ over all $A \subseteq V(G)$ is at most the $n/(2d)$ -th power of the sum of $q^{e(A)}$ over all subsets A of $V(K_{d,d})$. Carenini then asked for the exact extremal statement, which we restate in our notation.

Question ([2, Question 1.3]). *Suppose that $2d \mid n$, and let $m = n/(2d)$. Is it true that for every $\gamma \geq 0$, every d -regular graph G on n vertices satisfies $i_\gamma(G) \leq i_\gamma(mK_{d,d})$?*

For $\gamma = 0$ this is the theorem of Kahn and Zhao. For $\gamma \geq 1/2$ it holds trivially, since every set spans at most $dn/2 \leq \gamma dn$ edges and so $i_\gamma(G) = 2^n$. We call a pair (n, d) *admissible* if $d \geq 1$, $n \geq 1$ and $2d \mid n$, that is, if n is a positive multiple of $2d$. Seth’s count differs from i_γ , because its density condition depends on $|J|$; our results concern Question 1.3, and we make no claim about Seth’s count.

Results. For an integer t let $N_{\leq t}(G)$ be the number of sets $A \subseteq V(G)$ with $e_G(A) \leq t$. Since $e_G(A)$ is an integer, $i_\gamma(G) = N_{\leq \lfloor \gamma dn \rfloor}(G)$. Hence an inequality $N_{\leq t}(G) > N_{\leq t}(mK_{d,d})$ at an integer $t \geq 0$ gives a negative answer to Question 1.3 for every $\gamma \in [t/(dn), (t+1)/(dn))$. We write $T(G)$ for the number of triangles of G , C_k for the cycle of length k , and $G \square H$ for the Cartesian product.

Theorem A. *Let $d \geq 2$, let n be a positive multiple of $2d$ with $(n, d) \neq (4, 2)$, let $m = n/(2d)$, and put*

$$t^* = \frac{dn}{2} - 3d + 1, \quad \gamma^* = \frac{t^*}{dn} = \frac{1}{2} - \frac{3d-1}{dn}.$$

Then $t^ \geq 1$, and the following hold.*

- (a) *If $d \geq 3$, every d -regular graph G on n vertices satisfies $i_{\gamma^*}(G) - i_{\gamma^*}(mK_{d,d}) \geq \frac{7}{5}T(G)$, and $i_{\gamma^*}(G) = i_{\gamma^*}(mK_{d,d})$ if and only if G is triangle-free. So $mK_{d,d}$ minimises i_{γ^*} among d -regular graphs on n vertices, and every such graph that contains a triangle has a larger value.*
- (b) *If $d = 2$, every 2-regular graph $G \neq mC_4$ on n vertices satisfies $i_{\gamma^*}(G) > i_{\gamma^*}(mC_4)$.*

The excluded cases are trivial: for $d = 1$ the only 1-regular graph on n vertices is $\frac{n}{2}K_2 = \frac{n}{2}K_{1,1}$, and $C_4 = K_{2,2}$ is the only 2-regular graph on four vertices. For every other admissible pair there is a competitor. If $d \geq 3$, the graph $(K_d \square K_2) \cup (m-1)K_{d,d}$ is d -regular and contains a triangle; if $d = 2$ and $n \geq 8$, the cycle C_n differs from mC_4 . So the answer to Question 1.3 is negative for every admissible pair (n, d) except the trivial ones.

For $(n, d) = (6, 3)$ we get $t^* = 1$ and $\gamma^* = 1/18$. The only cubic graphs on six vertices are $K_{3,3}$ and the triangular prism $K_3 \square K_2$, and Table 1 lists their counts. The prism has

t	0	1	2	3	4	5	6	7	8	9
$N_{\leq t}(K_{3,3})$	15	24	42	48	57	57	63	63	63	64
$N_{\leq t}(K_3 \square K_2)$	13	28	40	48	57	57	63	63	63	64

Table 1: The number of vertex sets spanning at most t edges, for the two cubic graphs on six vertices. Here $i_\gamma = N_{\leq t}$ with $t = \lfloor 18\gamma \rfloor$.

28 sets spanning at most one edge, and $K_{3,3}$ has 24. So $K_{3,3}$ fails to be a maximiser exactly for $\gamma \in [1/18, 1/9)$. Since $n = 6$ is the smallest order at which Question 1.3 is not trivial, $(n, d, \gamma) = (6, 3, 1/18)$ is the smallest counterexample, in the sense that it has the smallest n and, for that n , the smallest γ .

For small γ we give two explicit families.

Theorem B. *Let $d \geq 3$, let $K_{d,d}$ have sides $X = \{x_1, \dots, x_d\}$ and $Y = \{y_1, \dots, y_d\}$, and let S_d be obtained from $K_{d,d}$ by deleting the edges x_1y_1 and x_2y_2 and adding the edges x_1x_2 and y_1y_2 . Then S_d is d -regular, every set spanning at most one edge of $K_{d,d}$ spans at most one edge of S_d , and*

$$N_{\leq 1}(S_d) = N_{\leq 1}(K_{d,d}) + 4d - 8 = 2^{d+1} + d^2 + 4d - 9.$$

Consequently, for $n = 2d$ and every $\gamma \in [1/(2d^2), 1/d^2)$, $K_{d,d}$ does not maximise i_γ among d -regular graphs on n vertices.

The graph S_3 is the prism. Corollary 5 gives an explicit criterion for $S_d \cup (m-1)K_{d,d}$ to beat $mK_{d,d}$.

Theorem B'. *Let $d \geq 2$. Take two disjoint copies (X_1, Y_1) and (X_2, Y_2) of $K_{d,d}$, choose $u_k \in X_k$ and $w_k \in Y_k$ for $k = 1, 2$, delete the edges u_1w_1 and u_2w_2 , and add the edges u_1w_2 and u_2w_1 . The resulting graph H_d is d -regular, bipartite and connected, and*

$$N_{\leq 1}(H_d) - N_{\leq 1}(2K_{d,d}) = 2(d-1)(2^d + d - 3) > 0, \quad i_0(H_d) - i_0(2K_{d,d}) = -2(2^{d-1} - 1)^2.$$

Consequently, for $n = 4d$ and every $\gamma \in [1/(4d^2), 1/(2d^2))$, $2K_{d,d}$ does not maximise i_γ , even among bipartite d -regular graphs on n vertices.

So the answer is negative even within the bipartite class for which Kahn proved the case $\gamma = 0$. The graph H_2 is the cycle C_8 .

At the exponential scale, on the other hand, the answer is positive for every γ . For $q > 0$ put

$$P_d(q) = \sum_{a,b=0}^d \binom{d}{a} \binom{d}{b} q^{ab},$$

which is the sum of $q^{e(A)}$ over all subsets A of $V(K_{d,d})$.

Proposition C. *Let $d \geq 2$ and $\gamma \geq 0$, and put*

$$\rho(d, \gamma) = \inf_{0 < q \leq 1} \left(\frac{1}{2d} \log P_d(q) - \gamma d \log q \right).$$

Every d -regular graph G on n vertices satisfies $\frac{1}{n} \log i_\gamma(G) \leq \rho(d, \gamma)$, and $\frac{1}{n} \log i_\gamma(mK_{d,d}) \rightarrow \rho(d, \gamma)$ as $n \rightarrow \infty$ through multiples of $2d$, where $m = n/(2d)$.

The upper bound is Carenini's Proposition 2.1 with the fugacity equal to 1, optimised over q . Proposition C shows that the maximum of i_γ over d -regular graphs on n vertices is $i_\gamma(mK_{d,d}) e^{o(n)}$ for fixed d and γ ; so every failure is subexponential. The failures are not confined to small n , however.

Theorem D. *Let $d \geq 2$ and $1/8 < \gamma < 1/2$. For n divisible by $2d$ write $n/(2d) = 2k + r$ with $r \in \{0, 1\}$, and let $G_n = kH_d \cup rK_{d,d}$, a bipartite d -regular graph on n vertices. Then $i_\gamma(G_n) > i_\gamma(mK_{d,d})$, where $m = n/(2d)$, for all sufficiently large n divisible by $2d$.*

The proof compares the upper tails of the two counts with Cramér’s theorem in its qualitative form, so it gives no explicit threshold n_0 beyond which the inequality holds.

Scope. In Theorems B and B’ the threshold is a single edge, $\lfloor \gamma dn \rfloor = 1$. In Theorem A, γ^* exceeds $1/8$ unless $(n, d) = (6, 3)$, and $\gamma^* \rightarrow 1/2$ as $n \rightarrow \infty$. In Theorem D, $\gamma > 1/8$ is fixed; both counts are then $(1 - o(1))2^n$, and their relative difference tends to 0 (Remark 10). The value $1/8$ is where the mean crosses the threshold: a uniformly random subset of a d -regular graph on n vertices spans $dn/8$ edges on average. Question 1.3 remains open for fixed $\gamma \in (0, 1/8]$ as $n \rightarrow \infty$. All our counterexamples with $\gamma \leq 1/8$ have $\gamma d < 1$: in Theorems B and B’ and Corollary 5 we have $\gamma < 1/d^2$, and the others, namely the case $(n, d) = (6, 3)$ of Theorem A, Remark 12 and the failures in Table 3, have $d \leq 5$. In particular, none of our counterexamples with $\gamma \leq 1/8$ has $\gamma d \geq 2$, the range of small γ in which Carenini’s Proposition 1.2 shows that the term $\gamma \log(e/\gamma)n$ of her Theorem 1.1 is needed. The weaker statement $\max_G i_\gamma(G) \leq (1 + o(1))i_\gamma(mK_{d,d})$ holds for $\gamma = 0$ and for every fixed $\gamma > 1/8$ (reading R6 in Section 2); it is open for $0 < \gamma < 1/8$, and we make no claim at $\gamma = 1/8$. Section 2 lists several readings of the question with our status for each, and Section 10 states open questions.

Prior work. Question 1.3 appeared in version 1 of [2], submitted to the arXiv on 22 September 2026. We found no earlier answer to it, and no discussion of it, in the following sources, which we searched on 26 September 2026 between 05:12 and 05:23 UTC: the arXiv (the listing of [2], which had only version 1, and searches of the arXiv API for later papers on the topic or by the same author); Semantic Scholar, which listed no citations of [2]; the problem database MathDB (<https://mathdb.com>); GitHub searches of code, commits, issues, pull requests and repositories; the public repositories of AI-assisted work on open problems that we check, among them the `trureturing` repository [13]; and web search. Some of these searches were repeated on 26 September 2026 between 06:39 and 06:42 UTC, with the same result: we searched the issues and pull requests of the `trureturing` repository for the versioned identifier 2609.28527v1 (as a positive control, the same search for the versioned identifier of another recent arXiv paper found the expected issue), searched GitHub issues and commits, and searched MathDB by title. For version 1 we did not search Google Scholar, zbMATH or MathOverflow. For version 2 we repeated these searches on 29 September 2026 between 12:55 and 13:35 UTC, this time also on MathOverflow and zbMATH; [2] still had only version 1, and we found no answer to the question other than version 1 of this paper. This reports the coverage of our searches, not a guarantee of priority. Theorem 1.1 and Proposition 2.1 of [2] are due to Carenini. The upper bound in Proposition C is a direct consequence of the latter, which rests on the theorem of Sah, Sawhney, Stoner and Zhao [11, Corollary 1.15].

Organisation. Section 2 discusses readings of the question. Section 3 collects notation and a complement identity. Theorem A is proved in Section 4, Theorems B and B’ in Section 5, Proposition C in Section 6 and Theorem D in Section 7. Section 8 reports computations, Section 9 the formal verification, and Section 10 open questions.

2 Readings of the question

Question 1.3 is a single universal statement, but it can be weakened in several natural ways. Table 2 lists the readings we considered; R1 is the literal one. In R4–R6, d and γ are fixed and

$n \rightarrow \infty$ through multiples of $2d$.

	Reading	Status
R1	Literal: for all admissible (n, d) and all $\gamma \geq 0$, $i_\gamma(G) \leq i_\gamma(mK_{d,d})$ for every d -regular graph G on n vertices.	False. Smallest counterexample (smallest n , then smallest γ): $(n, d, \gamma) = (6, 3, 1/18)$ (Theorems A and B). Both the counterexample and its minimality are formally verified.
R2	For a given admissible pair (n, d) , the inequality holds for all $\gamma \geq 0$.	False for every admissible pair with $d \geq 2$ and $(n, d) \neq (4, 2)$ (Theorem A); true in the trivial cases $d = 1$ and $(n, d) = (4, 2)$.
R3	Small γ : the inequality holds for all admissible (n, d) and all $0 \leq \gamma < 1/8$.	False: for $n = 2d$ and every $d \geq 3$ (Theorem B); for $n = 4d$ and every $d \geq 2$, with a connected bipartite competitor (Theorem B'); for some $n = 2dm$ with $m \geq 2$ (Corollary 5); at $\gamma = 1/10$ for many $n \leq 400$ (Remark 12).
R4	Fixed γ : the inequality holds for all large n .	False for every $d \geq 2$ and every $\gamma \in (1/8, 1/2)$ (Theorem D); true for $\gamma = 0$ and for $\gamma \geq 1/2$.
R5	R4 for fixed $\gamma \in (0, 1/8]$.	Open. True at the exponential scale (Proposition C). The competitors of Theorem D lose for large n when $\gamma < 1/8$ (Remark 11).
R6	Ratio form: $\max_G i_\gamma(G) \leq (1 + o(1)) i_\gamma(mK_{d,d})$.	Not refuted. True for $\gamma = 0$, and for $\gamma > 1/8$, where $i_\gamma(mK_{d,d}) = (1 - o(1))2^n$ (Remark 10). Open for $0 < \gamma < 1/8$; we make no claim at $\gamma = 1/8$.

Table 2: Readings of Question 1.3 of [2]; here $m = n/(2d)$.

3 Preliminaries

All graphs are finite and simple. For a graph G let $N_s(G)$ be the number of sets $A \subseteq V(G)$ with $e_G(A) = s$, and let

$$P_G(z) = \sum_{A \subseteq V(G)} z^{e_G(A)} = \sum_{s \geq 0} N_s(G) z^s,$$

so that $N_{\leq t}(G) = \sum_{s \leq t} [z^s] P_G(z)$ and $i_0(G) = N_0(G)$, where $[z^s] F(z)$ denotes the coefficient of z^s in a polynomial F . In the notation of [2], $P_G(q) = Z_G(1, q)$. For a disjoint union, $P_{G \cup F} = P_G P_F$, since $e_{G \cup F}(A) = e_G(A \cap V(G)) + e_F(A \cap V(F))$. Choosing a vertices on one side of $K_{d,d}$ and b on the other spans ab edges, so $P_{K_{d,d}} = P_d$ and $P_{mK_{d,d}} = P_d^m$. For a d -regular graph G on n vertices we write $E = e(G) = dn/2$. Every integer $t \geq 0$ arises as $\lfloor \gamma dn \rfloor$, namely for $\gamma \in [t/(dn), (t+1)/(dn))$, so Question 1.3 is equivalent to the inequalities $N_{\leq t}(G) \leq N_{\leq t}(mK_{d,d})$ for all integers $t \geq 0$.

Lemma 1 (complement identity). *Let G be a d -regular graph with n vertices and E edges, and for $S \subseteq V(G)$ let $\partial(S)$ be the number of edges meeting S . Then $\partial(S) = d|S| - e_G(S)$, and for every integer t ,*

$$N_{\leq t}(G) = 2^n - |\{S \subseteq V(G) : \partial(S) \leq E - t - 1\}|.$$

Proof. Summing the degrees of the vertices of S counts each edge inside S twice and each edge between S and its complement once, so $d|S| = e_G(S) + \partial(S)$. The edges inside $V(G) \setminus S$ are exactly the edges not meeting S , so $e_G(V(G) \setminus S) = E - \partial(S)$. Hence $e_G(A) \leq t$ if and only if $\partial(V(G) \setminus A) \geq E - t$, and $A \mapsto V(G) \setminus A$ is a bijection. $\square$

4 Proof of Theorem A

Proposition 2. *Let $d \geq 2$, let G be a d -regular graph on n vertices, and let $t^* = dn/2 - 3d + 1$. Then*

$$N_{\leq t^*}(G) = 2^n - 1 - n - \binom{n}{2} - n \binom{d}{2} + 2T(G) - Q(G),$$

where $Q(G)$ is the number of 4-sets spanning at least $d + 2$ edges plus the number of 5-sets spanning at least $2d + 2$ edges.

Proof. We have $E - t^* - 1 = 3d - 2$. By Lemma 1, $2^n - N_{\leq t^*}(G)$ is the number of sets S with $\partial(S) = d|S| - e_G(S) \leq 3d - 2$. We count these sets by their size k .

- $k \leq 2$: every set qualifies, since $\partial(S) \leq kd \leq 2d \leq 3d - 2$. This gives $1 + n + \binom{n}{2}$ sets.
- $k = 3$: the condition is $e_G(S) \geq 2$. Call a pair of edges with a common end a *cherry*; there are $n \binom{d}{2}$ cherries. A 3-set with exactly two edges contains exactly one cherry, and a triangle contains three. Hence there are $n \binom{d}{2} - 3T(G)$ sets of size 3 with exactly two edges, and $n \binom{d}{2} - 2T(G)$ with at least two.
- $k = 4$ and $k = 5$: the conditions are $e_G(S) \geq d + 2$ and $e_G(S) \geq 2d + 2$, so these sets are counted by $Q(G)$.
- $k \geq 6$: we would need $e_G(S) \geq (k - 3)d + 2$, while $2e_G(S) \leq kd$ as every vertex has degree d . Together these give $(k - 6)d \leq -4$, which is impossible.

Adding up proves the formula. $\square$

Lemma 3. *Let G be a d -regular graph. If $d \geq 5$ then $Q(G) = 0$; if $d = 4$ then $Q(G) \leq \frac{3}{5}T(G)$; if $d = 3$ then $Q(G) \leq \frac{1}{2}T(G)$; and if $d = 2$ then $Q(G)$ is the number $c_4(G)$ of components of G isomorphic to C_4 . In particular, if $d \geq 3$ then $2T(G) - Q(G) \geq \frac{7}{5}T(G)$.*

Proof. If $d \geq 5$, a 4-set spans at most $6 < d + 2$ edges and a 5-set spans at most $10 < 2d + 2$ edges.

Let $d = 4$. Then $Q(G)$ is the number of copies of K_4 plus the number of copies of K_5 . A triangle xyz lies in at most two copies of K_4 , because the fourth vertex is a neighbour of x other than y and z , and x has only two such neighbours. As each K_4 contains four triangles, the number of copies of K_4 is at most $2T(G)/4$. Each vertex of a K_4 has exactly one neighbour outside it, so a K_4 lies in at most one K_5 . As each K_5 contains five copies of K_4 , the number of copies of K_5 is at most a fifth of the number of copies of K_4 . Hence $Q(G) \leq \frac{6}{5} \cdot \frac{T(G)}{2} = \frac{3}{5}T(G)$.

Let $d = 3$. A 5-set spans at most $\lfloor 15/2 \rfloor = 7 < 8$ edges, so $Q(G)$ counts the 4-sets with at least five edges. Each of them contains at least two triangles. A triangle xyz lies in at most one of them: its fourth vertex must be adjacent to at least two of x, y, z ; if w and w' both have this property, then some vertex of $\{x, y, z\}$ is adjacent to both, and since that vertex has only one neighbour outside $\{x, y, z\}$, $w = w'$. Counting pairs consisting of a triangle and a set counted by $Q(G)$ that contains it gives $2Q(G) \leq T(G)$.

Let $d = 2$. A 4-set with four edges induces a 2-regular graph on four vertices, that is, a component C_4 ; and a 5-set spans at most five edges.

The last assertion follows because $\frac{3}{5} \geq \frac{1}{2}$. $\square$

Proof of Theorem A. First, $t^* \geq 1$: if $d \geq 3$, then $n \geq 2d$ gives $t^* \geq d^2 - 3d + 1 \geq 1$; if $d = 2$, then $n \geq 8$ and $t^* = n - 5 \geq 3$. As $\gamma^* dn = t^*$, we have $i_{\gamma^*} = N_{\leq t^*}$. By Proposition 2, for d -regular graphs G and G' on n vertices,

$$N_{\leq t^*}(G) - N_{\leq t^*}(G') = (2T(G) - Q(G)) - (2T(G') - Q(G')).$$

The graph $mK_{d,d}$ is bipartite, so $T(mK_{d,d}) = 0$. If $d \geq 3$, then also $Q(mK_{d,d}) = 0$ by Lemma 3, and (a) follows from Lemma 3: the difference $2T(G) - Q(G)$ is at least $\frac{7}{5}T(G)$, and it vanishes when $T(G) = 0$. If $d = 2$, Lemma 3 gives $N_{\leq t^*}(G) - N_{\leq t^*}(mC_4) = 2T(G) + n/4 - c_4(G)$, which is positive unless $T(G) = 0$ and $c_4(G) = n/4$, that is, unless $G = mC_4$. $\square$

Remark 4. For $n = 2d$, every d -regular graph $G \neq K_{d,d}$ contains a triangle. Indeed, if G is triangle-free and uv is an edge, the neighbourhoods $N(u)$ and $N(v)$ are disjoint independent sets of size d , so they partition $V(G)$; every vertex of $N(u)$ then has all its d neighbours in $N(v)$, and $G = K_{d,d}$. Hence, for every $d \geq 3$, every d -regular graph on $2d$ vertices other than $K_{d,d}$ beats $K_{d,d}$ at $t^* = d^2 - 3d + 1$. As a second example, for $(n, d) = (12, 3)$ we have $t^* = 10$ and $\gamma^* = 5/18$, and the graph $3K_4$ has $T = 12$ and $Q = 3$; so $N_{\leq 10}(3K_4) - N_{\leq 10}(2K_{3,3}) = 21$, the two counts being 4002 and 3981.

5 Small thresholds: proofs of Theorems B and B'

5.1 The switched complete bipartite graph

Proof of Theorem B. Write $K = K_{d,d}$. Each of x_1, x_2, y_1, y_2 loses one edge and gains one, so S_d is d -regular. For $A \subseteq X \cup Y$ put $A_X = A \cap X$, $A_Y = A \cap Y$, $a = |A_X|$ and $b = |A_Y|$, so that $e_K(A) = ab$.

Inclusion. If $e_K(A) \leq 1$, then $a = 0$, or $b = 0$, or $a = b = 1$. If $b = 0$, then $A \subseteq X$, and the only edge of S_d inside X is x_1x_2 ; the case $a = 0$ is symmetric; and a set of two vertices spans at most one edge. So $e_{S_d}(A) \leq 1$. The inclusion is strict: $A_0 = \{x_1, y_1, y_3\}$ has $e_K(A_0) = 2$, but $e_{S_d}(A_0) = 1$, since x_1y_1 was deleted and y_1y_3 is not an edge. (Here $d \geq 3$ is used.)

Exact count. By the inclusion, $N_{\leq 1}(S_d) - N_{\leq 1}(K)$ is the number of sets A with $e_K(A) \geq 2$ and $e_{S_d}(A) \leq 1$. Writing $[\cdot]$ for the indicator of a statement,

$$e_{S_d}(A) = ab - [x_1 \in A_X, y_1 \in A_Y] - [x_2 \in A_X, y_2 \in A_Y] + [x_1, x_2 \in A_X] + [y_1, y_2 \in A_Y].$$

We need $ab \geq 2$ and $e_{S_d}(A) \leq 1$. At most two terms are subtracted, so $ab \leq 3$.

- $(a, b) = (1, 2)$: we need one subtraction and no addition, that is, $A_X = \{x_1\}$ and $y_1 \in A_Y$, or $A_X = \{x_2\}$ and $y_2 \in A_Y$, and in both cases $A_Y \neq \{y_1, y_2\}$. The second vertex of A_Y is then one of $y_3, \dots, y_d$, so there are $2(d-2)$ such sets.
- $(a, b) = (2, 1)$: by symmetry, there are $2(d-2)$ such sets.
- $ab = 3$: then $a = 1$ or $b = 1$, at most one term is subtracted, and $e_{S_d}(A) \geq 2$.

Hence $N_{\leq 1}(S_d) - N_{\leq 1}(K) = 4d - 8$. Finally, $N_{\leq 1}(K) = 2^{d+1} - 1 + d^2$: the independent sets of K are the $2^{d+1} - 1$ subsets of X or of Y , and the sets spanning exactly one edge are the d^2 pairs $\{x, y\}$ with $x \in X$ and $y \in Y$. For $n = 2d$ and $\gamma \in [1/(2d^2), 1/d^2]$ we have $\lfloor \gamma dn \rfloor = 1$, which gives the last assertion. $\square$

Corollary 5. *Let $d \geq 3$ and $m \geq 2$, and put $I = 2^{d+1} - 1$. Then*

$$N_{\leq 1}(S_d \cup (m-1)K_{d,d}) - N_{\leq 1}(mK_{d,d}) = I^{m-2}[(4d-8)I - (m-1)d^2(2^{d-1} - 2)].$$

Hence, for $n = 2dm$ and $\gamma \in [1/(dn), 2/(dn))$, the graph $S_d \cup (m-1)K_{d,d}$ beats $mK_{d,d}$ if and only if $(4d-8)I > (m-1)d^2(2^{d-1} - 2)$. This holds exactly for $d = 3$ with $2 \leq m \leq 4$, for $d \in \{4, 5\}$ with $2 \leq m \leq 3$, and for $6 \leq d \leq 13$ with $m = 2$. For $d \geq 14$ it fails for every $m \geq 2$.

Proof. For disjoint graphs G and F , a set spans at most one edge of $G \cup F$ exactly when it spans at most one edge of G and none of F , or none of G and exactly one of F . Hence

$$N_{\leq 1}(G \cup F) = N_{\leq 1}(G) i_0(F) + i_0(G) (N_{\leq 1}(F) - i_0(F)).$$

Take $F = (m-1)K_{d,d}$, so that $i_0(F) = I^{m-1}$ and $N_{\leq 1}(F) - i_0(F) = (m-1)d^2 I^{m-2}$: one block spans one edge and the others none. Subtracting the identities for $G = S_d$ and $G = K_{d,d}$ and using Theorem B, it remains to show $i_0(K_{d,d}) - i_0(S_d) = 2^{d-1} - 2$, that is, $i_0(S_d) = 3 \cdot 2^{d-1} + 1$. An independent set of S_d inside X is a subset of X not containing both x_1 and x_2 , and there are $3 \cdot 2^{d-2}$ of these; the same holds for Y . An independent set meeting both sides has $a, b \geq 1$; if $a = b = 1$ it must be $\{x_1, y_1\}$ or $\{x_2, y_2\}$, and if $ab \geq 2$ then, by the formula for $e_{S_d}(A)$ above, $e_{S_d}(A) \geq ab - 1 \geq 1$ when $\min(a, b) = 1$, and $e_{S_d}(A) \geq ab - 2 \geq 2$ otherwise. So $i_0(S_d) = 2 \cdot 3 \cdot 2^{d-2} - 1 + 2 = 3 \cdot 2^{d-1} + 1$.

For the list, note that the right-hand side of the criterion increases with m , so for each d the values of m for which it holds form an interval starting at $m = 2$; for $3 \leq d \leq 13$ the list follows by direct evaluation. For $d \geq 14$, the right-hand side minus the left-hand side at $m = 2$ equals $(d^2 - 16d + 32)2^{d-1} - 2d^2 + 4d - 8$, which is positive because $d^2 - 16d + 32 = (d-8)^2 - 32 \geq 4$ and $2^{d+1} > 2d^2 + 8$. $\square$

5.2 A two-block switch

Let $K = K_{d,d}$ and, for $\alpha, \beta \in \{0, 1\}$,

$$W_{\alpha\beta}(z) = \sum_{i,j=0}^{d-1} \binom{d-1}{i} \binom{d-1}{j} z^{ij+\alpha j+\beta i}.$$

Exchanging i and j shows $W_{10} = W_{01}$. Put $D(z) = W_{00}(z)W_{11}(z) - W_{10}(z)^2$.

Lemma 6. *Let $d \geq 2$ and let H_d be as in Theorem B'. Then*

$$P_{2K}(z) - P_{H_d}(z) = 2(z-1)D(z).$$

Consequently, for every graph F , possibly with no vertices, and every integer $t \geq 0$,

$$N_{\leq t}(H_d \cup F) - N_{\leq t}(2K \cup F) = 2[z^t](D(z)P_F(z)).$$

Proof. The graphs $2K$ and H_d have the same vertex set. For A in it and $k = 1, 2$ let $\alpha_k = [u_k \in A]$, $\beta_k = [w_k \in A]$, $i_k = |A \cap (X_k \setminus \{u_k\})|$ and $j_k = |A \cap (Y_k \setminus \{w_k\})|$. The edges of the k -th copy of K other than $u_k w_k$ that lie inside A number $i_k j_k + \alpha_k j_k + \beta_k i_k$, and both graphs contain all of them. In addition, $2K$ contains the edges $u_1 w_1$ and $u_2 w_2$, which contribute $\alpha_1 \beta_1 + \alpha_2 \beta_2$, while H_d contains $u_1 w_2$ and $u_2 w_1$, which contribute $\alpha_1 \beta_2 + \alpha_2 \beta_1$. Summing over A with $(\alpha_1, \beta_1, \alpha_2, \beta_2)$ fixed,

$$P_{2K} = \sum W_{\alpha_1 \beta_1} W_{\alpha_2 \beta_2} z^{\alpha_1 \beta_1 + \alpha_2 \beta_2}, \quad P_{H_d} = \sum W_{\alpha_1 \beta_1} W_{\alpha_2 \beta_2} z^{\alpha_1 \beta_2 + \alpha_2 \beta_1},$$

the sums being over $\{0, 1\}^4$. The two exponents differ by $(\alpha_1 - \alpha_2)(\beta_1 - \beta_2)$, which vanishes unless $\alpha_1 \neq \alpha_2$ and $\beta_1 \neq \beta_2$. The patterns $(\alpha_1, \beta_1, \alpha_2, \beta_2) = (1, 1, 0, 0)$ and $(0, 0, 1, 1)$ each contribute $(z-1)W_{00}W_{11}$ to $P_{2K} - P_{H_d}$, and the patterns $(1, 0, 0, 1)$ and $(0, 1, 1, 0)$ each contribute $(1-z)W_{10}W_{01}$. As $W_{10} = W_{01}$, this proves the identity. For the second assertion, put $R = DP_F$. Since $P_{G \cup F} = P_G P_F$,

$$\begin{aligned} N_{\leq t}(H_d \cup F) - N_{\leq t}(2K \cup F) &= -2 \sum_{s \leq t} [z^s]((z-1)R) \\ &= -2 \sum_{s \leq t} ([z^{s-1}]R - [z^s]R) = 2[z^t]R. \end{aligned} \quad \square$$

Lemma 7. *Let $d \geq 2$ and $q > 0$. Then $D(q) > 0$ if $q > 1$, and $D(q) < 0$ if $q < 1$.*

Proof. For $i, j \in \{0, \dots, d-1\}$ put $\mu(i, j) = \binom{d-1}{i} \binom{d-1}{j} q^{ij}$, and consider

$$\Delta = \sum_{i, i', j, j'} [\mu(i, j) \mu(i', j') - \mu(i, j') \mu(i', j)] (q^i - q^{i'}) (q^j - q^{j'}),$$

where all four indices range over $\{0, \dots, d-1\}$. We have $W_{00}(q) = \sum \mu(i, j)$, $W_{11}(q) = \sum \mu(i, j) q^{i+j}$ and $W_{10}(q) = \sum \mu(i, j) q^j = \sum \mu(i, j) q^i$. Expanding $(q^i - q^{i'}) (q^j - q^{j'})$ into four terms, we find

$$\begin{aligned} \sum_{i, i', j, j'} \mu(i, j) \mu(i', j') (q^i - q^{i'}) (q^j - q^{j'}) &= 2W_{00}W_{11} - 2W_{10}^2, \\ \sum_{i, i', j, j'} \mu(i, j') \mu(i', j) (q^i - q^{i'}) (q^j - q^{j'}) &= 2W_{10}^2 - 2W_{00}W_{11}, \end{aligned}$$

all at the point q . Subtracting, $\Delta = 4D(q)$.

Now let $q \neq 1$. The bracket equals $\binom{d-1}{i} \binom{d-1}{i'} \binom{d-1}{j} \binom{d-1}{j'} q^{ij'+i'j} (q^{(i-i')(j-j')} - 1)$, which has the sign of $(i-i')(j-j') \log q$. Since q^x is strictly monotone in x , the product $(q^i - q^{i'}) (q^j - q^{j'})$ has the sign of $(i-i')(j-j')$. So every term of Δ is zero or has the sign of $\log q$, and the term with $i = j = 0$ and $i' = j' = 1$, which exists because $d-1 \geq 1$, is not zero. $\square$

Proof of Theorem B'. Each of u_1, u_2, w_1, w_2 loses one edge and gains one, so H_d is d -regular. It is bipartite with sides $X_1 \cup X_2$ and $Y_1 \cup Y_2$, since the new edges join X_1 to Y_2 and X_2 to Y_1 . A copy of $K_{d,d}$ minus one edge is connected when $d \geq 2$, and the edge $u_1 w_2$ joins the two copies, so H_d is connected.

By Lemma 6 with F the graph with no vertices ($P_F = 1$), $N_{\leq t}(H_d) - N_{\leq t}(2K_{d,d}) = 2[z^t]D$ for every $t \geq 0$. The exponent in W_{00} is ij , so $[z^0]W_{00} = 2^d - 1$ (the pairs with $i = 0$ or $j = 0$) and $[z^1]W_{00} = (d-1)^2$. The exponent in W_{11} is $(i+1)(j+1) - 1$, so $[z^0]W_{11} = 1$ and $[z^1]W_{11} = 2(d-1)$. The exponent in W_{10} is $j(i+1)$, so $[z^0]W_{10} = 2^{d-1}$ and $[z^1]W_{10} = d-1$. Hence

$$\begin{aligned} [z^0]D &= 2^d - 1 - 4^{d-1} = -(2^{d-1} - 1)^2, \\ [z^1]D &= 2(d-1)(2^d - 1) + (d-1)^2 - 2 \cdot 2^{d-1}(d-1) = (d-1)(2^d + d - 3). \end{aligned}$$

Since $i_0 = N_{\leq 0}$, this proves the two formulas. For $n = 4d$ and $\gamma \in [1/(4d^2), 1/(2d^2))$ we have $\lfloor \gamma dn \rfloor = 1$. $\square$

6 The exponential scale: proof of Proposition C

Let ξ be the number of edges of $K_{d,d}$ spanned by a uniformly random subset of $V(K_{d,d})$. Then ξ takes values in $\{0, \dots, d^2\}$, $\mathbb{P}(\xi = 0) > 0$, $\mathbb{P}(\xi = d^2) = 4^{-d}$, $\mathbb{E}\xi = d^2/4$ because each edge is spanned with probability $1/4$, and $\mathbb{E}q^\xi = 4^{-d}P_d(q)$ for $q > 0$. For $m \geq 1$ let $\Sigma_m = \xi_1 + \dots + \xi_m$, where $\xi_1, \dots, \xi_m$ are independent copies of ξ . A uniformly random subset of $V(mK_{d,d})$ meets the m copies in independent uniformly random subsets. Hence, for $n = 2dm$ and $x = 2\gamma d^2$, so that $\gamma dn = mx$,

$$i_\gamma(mK_{d,d}) = 2^n \mathbb{P}(\Sigma_m \leq mx), \quad 2^n - i_\gamma(mK_{d,d}) = 2^n \mathbb{P}(\Sigma_m > mx). \quad (1)$$

We use the lower bound in Cramér's theorem in the following form.

Lemma 8. *Let ξ be a random variable with values in $\{0, 1, \dots, M\}$ such that $\mathbb{P}(\xi = 0) > 0$ and $\mathbb{P}(\xi = M) > 0$, let Σ_m be the sum of m independent copies of ξ , and let $\Lambda(\lambda) = \log \mathbb{E}e^{\lambda\xi}$.*

(a) *If $\mathbb{E}\xi < x < M$, then $\liminf_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{P}(\Sigma_m > mx) \geq \inf_{\lambda \geq 0} (\Lambda(\lambda) - \lambda x)$.*

(b) If $0 < x < \mathbb{E}\xi$, then $\liminf_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{P}(\Sigma_m < mx) \geq \inf_{\lambda \leq 0} (\Lambda(\lambda) - \lambda x)$.

Proof. Let $\Lambda^*(y) = \sup_{\lambda \in \mathbb{R}} (\lambda y - \Lambda(\lambda))$. By Cramér's theorem [3, Theorem 2.2.3], for every open set $U \subseteq \mathbb{R}$,

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{P}(\Sigma_m/m \in U) \geq - \inf_{y \in U} \Lambda^*(y).$$

The function Λ^* is convex, as a supremum of affine functions. It is finite on $[0, M]$: for all λ we have $\Lambda(\lambda) \geq \lambda M + \log \mathbb{P}(\xi = M)$ and $\Lambda(\lambda) \geq \log \mathbb{P}(\xi = 0)$, so for $0 \leq y \leq M$ the quantity $\lambda y - \Lambda(\lambda)$ is at most $-\log \mathbb{P}(\xi = M)$ when $\lambda \geq 0$ and at most $-\log \mathbb{P}(\xi = 0)$ when $\lambda \leq 0$. Hence Λ^* is continuous on $(0, M)$. By Jensen's inequality $\Lambda(\lambda) \geq \lambda \mathbb{E}\xi$, so for $y \geq \mathbb{E}\xi$ the terms with $\lambda < 0$ are at most 0, the value at $\lambda = 0$; thus $\Lambda^*(y) = \sup_{\lambda \geq 0} (\lambda y - \Lambda(\lambda))$, which is nondecreasing in $y \geq \mathbb{E}\xi$. In case (a), take $U = (x, \infty)$; then $\inf_{y > x} \Lambda^*(y) = \lim_{y \downarrow x} \Lambda^*(y) = \Lambda^*(x) = - \inf_{\lambda \geq 0} (\Lambda(\lambda) - \lambda x)$. Case (b) is symmetric, with $U = (-\infty, x)$ and Λ^* nonincreasing on $(-\infty, \mathbb{E}\xi]$. $\square$

Proof of Proposition C. Upper bound. By [2, Proposition 2.1] with $\lambda = 1$, every d -regular graph G on n vertices satisfies $i_\gamma(G) \leq e^{\gamma t n} P_d(e^{-t/d})^{n/(2d)}$ for every $t \geq 0$. With $q = e^{-t/d} \in (0, 1]$ this reads $i_\gamma(G) \leq q^{-\gamma d n} P_d(q)^{n/(2d)}$. Taking logarithms, dividing by n and taking the infimum over q gives $\frac{1}{n} \log i_\gamma(G) \leq \rho(d, \gamma)$. (The proof in [2] combines the bound $P_G(q) \leq P_d(q)^{n/(2d)}$ of [11, Corollary 1.15] with the inequality $[e_G(A) \leq \gamma d n] \leq q^{e_G(A) - \gamma d n}$.)

Lower bound. Let $n = 2dm$, and let $x = 2\gamma d^2$ and Σ_m be as in (1). By the upper bound, it suffices to show that the lower limit of $\frac{1}{n} \log i_\gamma(mK_{d,d})$ as $m \rightarrow \infty$ is at least $\rho(d, \gamma)$.

- $\gamma = 0$: here $i_0(mK_{d,d}) = i_0(K_{d,d})^m$ exactly, so $\frac{1}{n} \log i_0(mK_{d,d}) = \frac{1}{2d} \log i_0(K_{d,d}) = \lim_{q \rightarrow 0} \frac{1}{2d} \log P_d(q) \geq \rho(d, 0)$.
- $0 < \gamma < 1/8$: then $0 < x < d^2/4 = \mathbb{E}\xi$. By (1) and Lemma 8(b), with $q = e^\lambda$,

$$\begin{aligned} \liminf_{m \rightarrow \infty} \frac{1}{m} \log i_\gamma(mK_{d,d}) &\geq 2d \log 2 + \inf_{0 < q \leq 1} (\log(4^{-d} P_d(q)) - x \log q) \\ &= \inf_{0 < q \leq 1} (\log P_d(q) - x \log q). \end{aligned}$$

Dividing by $2d$ and noting $x/(2d) = \gamma d$ gives the claim.

- $\gamma \geq 1/8$: then $x \geq \mathbb{E}\xi$, and $\mathbb{P}(\Sigma_m \leq mx) \geq \mathbb{P}(\Sigma_m \leq m\mathbb{E}\xi) \rightarrow 1/2$ by the central limit theorem, as ξ is not constant. Hence $\frac{1}{n} \log i_\gamma(mK_{d,d}) \rightarrow \log 2$, and $\log 2 \geq \rho(d, \gamma)$ is the value at $q = 1$. $\square$

Remark 9. In the notation of [2], $P_d(e^{-t/d}) = B_d(t)$, so $\rho(d, \gamma) = \inf_{t \geq 0} (\gamma t + \frac{1}{2d} \log B_d(t))$. Carenini's proof of Theorem 1.1 bounds this infimum by choosing $t = 2 \log(e/\gamma)$.

7 Fixed thresholds above one eighth: proof of Theorem D

Let $n = 2dm$, $x = 2\gamma d^2$ and ξ, Σ_m as in Section 6; for $1/8 < \gamma < 1/2$ we have $d^2/4 < x < d^2$. For a graph G on n vertices let $U(G) = 2^n - i_\gamma(G)$, the number of sets spanning more than $\gamma d n$ edges. We must show $U(G_n) < U(mK_{d,d})$ for all large m . Write $H = H_d$.

Proof of Theorem D. Lower bound for $U(mK_{d,d})$. Let $\varphi(q) = \log P_d(q) - x \log q$ for $q > 0$. Since $P_d(q) = 4^d \mathbb{E} q^\xi$, we have $\varphi'(1) = \mathbb{E}\xi - x < 0$; and $\varphi(q) \geq (d^2 - x) \log q \rightarrow \infty$ as $q \rightarrow \infty$, because $P_d(q) \geq q^{d^2}$. Hence φ attains its minimum over $[1, \infty)$ at some $q^* > 1$. By (1) and Lemma 8(a), with $q = e^\lambda$, for every $\varepsilon > 0$ and all large m ,

$$U(mK_{d,d}) = 2^n \mathbb{P}(\Sigma_m > mx) \geq 2^n \exp(m(\varphi(q^*) - d \log 4 - \varepsilon)) = P_d(q^*)^m (q^*)^{-mx} e^{-\varepsilon m}.$$

Upper bound for $U(G_n)$. For $q \geq 1$ and every set A we have $[e(A) > \gamma dn] \leq q^{e(A) - \gamma dn}$. Summing over A with $q = q^*$ gives $U(G_n) \leq (q^*)^{-mx} P_{G_n}(q^*) = (q^*)^{-mx} P_H(q^*)^k P_d(q^*)^r$.

Comparison. As $m - r = 2k$,

$$\frac{U(mK_{d,d})}{U(G_n)} \geq \left(\frac{P_d(q^*)^2}{P_H(q^*)} \right)^k e^{-\varepsilon m}.$$

By Lemma 6 at $z = q^*$ and Lemma 7, $P_d(q^*)^2 - P_H(q^*) = 2(q^* - 1)D(q^*) > 0$, so $\beta = \log(P_d(q^*)^2/P_H(q^*)) > 0$. Take $\varepsilon = \beta/4$. Since $k \geq (m - 1)/2$, the ratio is at least $\exp(\beta(m - 1)/2 - \beta m/4) = \exp(\beta(m - 2)/4) > 1$ for all large m . Hence $U(G_n) < U(mK_{d,d})$, that is, $i_\gamma(G_n) > i_\gamma(mK_{d,d})$, for all large n divisible by $2d$. $\square$

Remark 10. For $\gamma > 1/8$ the law of large numbers gives $\mathbb{P}(\Sigma_m \leq mx) \rightarrow 1$, so $i_\gamma(mK_{d,d}) = (1 - o(1))2^n$, and the relative difference $i_\gamma(G_n)/i_\gamma(mK_{d,d}) - 1 \leq U(mK_{d,d})/i_\gamma(mK_{d,d})$ tends to 0 exponentially fast. For example, for $d = 2$, $\gamma = 1/4$ and $n = 400$ the relative difference is about $1.3 \cdot 10^{-17}$. So the competitors of Theorem D win only by a factor $1 + o(1)$. The base $P_d(q^*)^2/P_H(q^*)$ tends to 1 as $\gamma \downarrow 1/8$, since then $q^* \rightarrow 1$; for $d = 2$ it is about 1.0006 at $\gamma = 1/4$ and 1.0001 at $\gamma = 0.49$. Our proof gives no explicit n_0 , and we give no value.

The value $1/8$ comes from the mean. Let A be a uniformly random subset of $V(G)$, where G is d -regular on n vertices. Then $e_G(A)$ is the sum of the indicators $[u, v \in A]$ over the $dn/2$ edges uv of G ; it has mean $dn/8$ and variance $dn(2d+1)/32$, as each indicator has variance $3/16$ and each of the $nd(d-1)$ ordered pairs of distinct edges with a common end has covariance $1/16$.

Remark 11. For $0 < \gamma < 1/8$ the same argument, applied to the lower tail, shows that G_n loses to $mK_{d,d}$ for all large n . Indeed, now $0 < x < \mathbb{E}\xi$, so φ is minimised over $(0, 1]$ at some $q^* < 1$ (note $\varphi'(1) > 0$ and $\varphi(q) \rightarrow \infty$ as $q \rightarrow 0$). By Lemma 7, $D(q^*) < 0$, so again $P_d(q^*)^2 - P_H(q^*) = 2(q^* - 1)D(q^*) > 0$. Lemma 8(b) gives $i_\gamma(mK_{d,d}) \geq P_d(q^*)^m (q^*)^{-mx} e^{-\varepsilon m}$, the inequality $[e(A) \leq \gamma dn] \leq q^{e(A) - \gamma dn}$ for $q \leq 1$ gives $i_\gamma(G_n) \leq (q^*)^{-mx} P_H(q^*)^k P_d(q^*)^r$, and the comparison is as above.

Remark 12. The small- γ failures are nevertheless not confined to tiny graphs. At $\gamma = 1/10$, the graph $G_n = kH_d \cup rK_{d,d}$ of Theorem D beats $mK_{d,d}$ for 92 of the 119 values $n \in \{8, 12, \dots, 480\}$ when $d = 2$, the largest being $n = 400$, and for 56 of the 119 values $n \in \{12, 18, \dots, 720\}$ when $d = 3$, the largest being $n = 360$ (Section 8). By Remark 11 these are finite-size effects.

8 Computations

In version 2 every mathematical claim of this section is proved in Lean (Section 9). The programs described here were written for version 1 and remain independent cross-checks. All counts in this paper were computed in exact integer arithmetic. Our programs compute the distribution $(N_s(G))_s$ in two independent ways: by a recursion over bitmasks, $e_G(A \cup \{v\}) = e_G(A) + |N(v) \cap A|$ for a vertex v larger than every element of A , and by a direct loop over all sets and all edges. The two agree on eleven test graphs. Counts for disjoint unions are obtained from $P_{G \cup F} = P_G P_F$. Two referee agents of version 1 (see the disclosure at the end of the paper) recomputed the main numbers, each with its own programs. Our programs and their logs are available in version 1.3.0 of the repository [4] (directory `certificates/carenini/`).

Exhaustive search at $n = 2d$. For $d \in \{3, 4, 5\}$ we generated all d -regular graphs on $2d$ vertices up to isomorphism, starting from $K_{d,d}$ and applying 2-switches (replacing two edges ab, cd by ac, bd or by ad, bc when these are not edges), with exact isomorphism tests. Any two graphs on the same vertex set with the same degree at every vertex are connected by a sequence of 2-switches [7, 6], so this finds every class. We found 2, 6 and 60 classes. Every d -regular

graph on $2d$ vertices is connected, since each component has at least $d+1$ vertices, and these are the numbers of connected cubic graphs on 6 vertices, quartic graphs on 8 vertices and quintic graphs on 10 vertices (sequences A002851, A006820 and A006821 of [10]). For $d = 3$ the first referee agent of version 1 also enumerated all 70 labelled cubic graphs on six vertices. Table 3 lists the thresholds at which $K_{d,d}$ is not a maximiser.

d	n	classes	t	$\max_G N_{\leq t}(G)$	$N_{\leq t}(K_{d,d})$
3	6	2	1	28	24
4	8	6	1	61	47
			3	137	127
			5	187	171
5	10	60	1	116	88
			2	203	188
			3	321	288
			5	524	448
			7	679	648
			8	783	748
			11	908	868

Table 3: All thresholds t at which $K_{d,d}$ does not maximise $N_{\leq t}$ among d -regular graphs on $n = 2d$ vertices, for $3 \leq d \leq 5$. The maximum is over all such graphs; the failure holds for $\gamma \in [t/(2d^2), (t+1)/(2d^2))$.

At every other threshold $t \leq d^2$, $K_{d,d}$ is a maximiser. It is the unique maximiser for $t \in \{0, 2\}$ when $d = 3$, for $t \in \{0, 4, 6\}$ when $d = 4$, and for $t \in \{0, 4, 6, 9, 10, 12\}$ when $d = 5$; at the remaining thresholds it ties with other graphs. (By Lemma 1, for $d \geq 3$ all d -regular graphs on n vertices have the same value of $N_{\leq t}$ when $t \geq dn/2 - 2d$.) Theorem B gives the failures at $t = 1$, and Theorem A with Remark 4 gives those at $t = d^2 - 3d + 1$, that is, at $t = 1, 5, 11$.

Other checks. The formula of Proposition 2 was checked on 100 regular graphs with $2 \leq d \leq 8$ and $n \leq 20$ (unions of copies of $K_{d,d}$, $(K_d \square K_2) \cup (m-1)K_{d,d}$, cycles, $C_3 \cup C_{n-3}$, unions of copies of K_{d+1} , and random regular graphs), with no mismatch; the first referee agent of version 1 checked 75 further graphs, including graphs in which 5-sets contribute to Q . Theorem B was checked for $3 \leq d \leq 12$, and Corollary 5 against brute force for $(d, m) \in \{(3, 2), (3, 3), (4, 2), (5, 2)\}$. The identity of Lemma 6 was checked symbolically for $2 \leq d \leq 5$, and the difference in Theorem B' for $2 \leq d \leq 20$; the sign in Lemma 7 was checked on a grid of values of q , as a sanity check. For Theorem D, we compared kH_d with $2kK_{d,d}$ exactly for $d \in \{2, 3, 4\}$ and all $n \leq 600$ divisible by $4d$. At $n = 192$ the competitor wins for $\gamma \in \{1/5, 1/4, 1/3, 2/5, 9/20\}$, while for $\gamma = 3/20$, closer to $1/8$, it still loses. The outcome is not monotone in n , so this does not show that $n_0 \leq 192$: for $d = 4$ and $\gamma = 9/20$ the competitor loses at $n = 128$ and again at $n = 208$, and for $d = 3$ and $\gamma = 9/20$ it wins at $n = 96$ and $n = 192$ but loses at $n = 144$. For $\gamma = 3/20$ it wins for every such n from 360 to 600 when $d = 2$, from 336 to 600 when $d = 3$, and from 320 to 592 when $d = 4$. The counts in Remark 12 are exact comparisons for all n in the stated ranges.

9 Formal verification

All the results are formally verified in Lean 4 (version 4.34.1) with Mathlib [9] (version v4.34.1, commit d13f23b7). Version 1 of this paper was released with a partial formalisation, the module `Carenini.lean` of version 1.3.0 of [4], which proved the five negative instances of Table 4. For

version 2 every result of the paper was formalised: each of its mathematical claims is a statement of Table 5, a definition, a cited result or a step of a written proof.

Statements before formal proofs. Here the statements were written after the written proofs: they formalise a paper that had already been released. On 27 September 2026, after version 1 had been released, an agent wrote a Lean file of definitions and statements, the *statement file*, which proves none of its statements. An independent agent reviewed it against the paper and the cited sources and checked its definitions with programs of its own; it was frozen by recording its SHA-256 hash `5cf68b68181d0528bab5ebc8a2b18abe29b0409b1f40d23f3176d3ed45873163` before any proof was written, and a second AI system then audited the frozen file. Version 2 of the file leaves out the two claims removed from this version of the paper (parts of three statements), adds two statements (the negation of Question 1.3 as a single statement, and the statement that our counterexamples with $\gamma \leq 1/8$ have $\gamma d < 1$), and adds the values $t^* = 1$ and $\gamma^* = 1/18$ at $(n, d) = (6, 3)$ to an existing statement (GammaStarScope); the other 54 statements are unchanged. An independent agent reviewed version 2; after two edits to comments that the review suggested, it was frozen on 29 September 2026 at 01:31 UTC by recording its SHA-256 hash `846efb56558f734ba475092854effbe7598549b68e6ea8b7bcad19e9c02d8777`. The proofs were written in separate files that import the statement file, which was not changed; work on proofs of statements unchanged from version 1 began at 01:18 UTC, before version 2 was frozen. The definitions of $e_G(A)$, $N_{\leq t}$, i_γ and of the question are copied verbatim from the module of version 1, and $mK_{d,d}$ is defined from Mathlib’s Cartesian product and complete bipartite graph.

What is proved. The statement file states 60 propositions, listed in Table 5; the file of checks (`Research/Backfill/Paper3/Check.lean`) proves each of them, by a theorem `check_X : X` for every name `X` of the table. The statement file also defines i_γ , the question and its literal form; in ASCII notation (`forall`, `->`, `<=`, `Not`, `Dvd.dvd a b for a | b`, `Nat` and `Real` for $\mathbb{N}$ and $\mathbb{R}$, and `g` for γ ; the directive `open Classical` in before `iGamma` is omitted),

```
noncomputable def iGamma (G : SimpleGraph V) [DecidableRel G.Adj] (d : Nat) (g : Real) : Nat :=
  #((univ : Finset (Finset V)).filter
    (fun A => ((edgesIn G A : Nat) : Real) <= g * d * (Fintype.card V : Real)))
def CareniniQuestion (n d : Nat) (g : Real) : Prop :=
  forall (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V)
    [DecidableRel G.Adj], Fintype.card V = n -> G.IsRegularOfDegree d ->
    iGamma G d g <= iGamma (KddUnion (n / (2 * d)) d) d g
def Question13 : Prop :=
  forall n d : Nat, 1 <= d -> Dvd.dvd (2 * d) n -> forall g : Real, 0 <= g ->
    CareniniQuestion n d g
def AnswerNegative : Prop := Not Question13
```

Here `edgesIn G A` is the number of edges of `G` (elements of Mathlib’s `G.edgeFinset`) with both ends in `A`, so `iGamma` is $i_\gamma(G)$ with the threshold γdn compared in $\mathbb{R}$, and `KddUnion m d` is $mK_{d,d}$. The predicate quantifies over every finite vertex type with n elements and every d -regular simple graph on it (Mathlib’s `IsRegularOfDegree`).

(n, d, γ)	competitor	counts $N_{\leq t}$	Lean statement
$(6, 3, 1/18)$	$\text{prism } K_3 \square K_2$	$28 > 24$ ($t = 1$)	Neg_6_3
$(2d, d, 1/(2d^2)), d \geq 3$	S_d	Theorem B ($t = 1$)	Neg_2d
$(8, 2, 1/16)$	$H_2 = C_8$	$111 > 105$ ($t = 1$)	Neg_8_2
$(12, 3, 1/36)$	H_3	$527 > 495$ ($t = 1$)	Neg_12_3_bip
$(12, 3, 5/18)$	$3K_4$	$4002 > 3981$ ($t = 10$)	Neg_12_3_top

Table 4: The five negative instances formalised in version 1. In each case the reference graph is $\frac{n}{2d}K_{d,d}$; the eight counts form the statement Table4Counts.

Table 5: The 60 statements of the frozen statement file (namespace BackfillPaper3.Challenge, version 2). Each is proved. The names are those of the statement file: Table2 formalises Table 3, and Table4Counts the counts of Table 4.

Lean statement	what it says	here
IGammaFloor	$i_\gamma = N_{\leq \lfloor \gamma dn \rfloor}$ for $\gamma \geq 0$	§1
IGammaIso	i_γ is invariant under isomorphism, for $\gamma \geq 0$	used throughout
PolyFacts	$P_{G \cup F} = P_G P_F$, $P_{K_{d,d}} = P_d$, $P_{mK_{d,d}} = P_d^m$ and $N_{\leq t}(G) = \sum_{s \leq t} [z^s] P_G$	§3
IntegerForm	for admissible (n, d) , Question 1.3 for all $\gamma \geq 0$ is equivalent to $N_{\leq t}(G) \leq N_{\leq t}(mK_{d,d})$ for all $t \geq 0$	§3
Neg_6_3, Neg_2d, Neg_8_2, Neg_12_3_bip, Neg_12_3_top	the five negative instances of Table 4	Table 4
Table4Counts	the eight counts of Table 4	Table 4
KddIndependentSets	$K_{d,d}$ has $2^{d+1} - 1$ independent sets	§1
Lemma1	Lemma 1	§3
Proposition2	Proposition 2	§4
Lemma3	Lemma 3	§4
TheoremA	Theorem A, with $t^* \geq 1$ and $\gamma^* = \frac{1}{2} - \frac{3d-1}{dn}$	§1, §4
Remark4	Remark 4	§4
TheoremB	Theorem B	§1, §5
Corollary5	Corollary 5, with $i_0(S_d) = 3 \cdot 2^{d-1} + 1$	§5
WSymm	$W_{10} = W_{01}$	§5
Lemma6	Lemma 6	§5
Lemma7	Lemma 7	§5
TheoremBprime	Theorem B'	§1, §5
Lemma8	Lemma 8, for independent copies of ξ on any probability space	§6
Lemma8_bddBelow	the two infima in Lemma 8 are over sets bounded below	well-definedness
PropositionC	Proposition C	§1, §6
Rho_bddBelow	the infimum defining $\rho(d, \gamma)$ is over a set bounded below	well-definedness
TheoremD	Theorem D, with the facts that G_n is a bipartite d -regular graph on n vertices	§1, §7
SSSZBound	$P_G(q) \leq P_d(q)^{n/(2d)}$ for d -regular G on n vertices and $0 < q \leq 1$, the case of [11, Corollary 1.15] used in [2]	§1
CareniniProp21	[2, Proposition 2.1] with fugacity 1	§6
KahnZhao	$i_0(G) \leq (2^{d+1} - 1)^{n/(2d)}$ for d -regular G (Kahn, Zhao)	§1
QuestionTrueHalf	the answer is yes for $\gamma \geq 1/2$	§1, R4
QuestionTrueZero	the answer is yes for $\gamma = 0$	§1, R4
TrivialCasesUnique	for $d = 1$ and for $(n, d) = (4, 2)$ the only d -regular graph is $mK_{d,d}$	§1
TrivialCasesTrue	in these cases the answer is yes for every γ	R2
CompetitorFacts	the competitors of §1 exist: $(K_d \square K_2) \cup (m-1)K_{d,d}$ is d -regular with a triangle, and $C_n \neq mC_4$ for $n \geq 8$; $C_4 = K_{2,2}$	§1
NegativeEveryAdmissible	the answer is no at $\gamma = \gamma^* \in (0, 1/2)$ for every admissible pair other than the trivial ones	§1, R2

Lean statement	what it says	here
AnswerNegative	the answer to Question 1.3 is no	abstract, R1
GammaStarScope	$\gamma^* > 1/8$ unless $(n, d) = (6, 3)$, $\gamma^* \rightarrow 1/2$, and $t^* = 1$, $\gamma^* = 1/18$ at $(6, 3)$	§1
LowGammaExamples	$\gamma < 1/d^2$ implies $\gamma d < 1$, and $d \leq 5$, $\gamma \leq 1/8$ imply $\gamma d < 1$; together these cover our counterexamples with $\gamma \leq 1/8$	§1, §10
S3PrismH2Cycle	S_3 is the prism and H_2 is C_8	§1
SethDensity	Seth's density condition implies $e_G(J) \leq dn/(2k)$	§1
MaximisersAtGammaStar	at γ^* the maximisers of i_{γ^*} are the maximisers of $2T(G) - Q(G)$	§10
Table1	Table 1	§1
CubicSix	the only cubic graphs on six vertices are $K_{3,3}$ and the prism	§1
ExactFailureSix	at $(n, d) = (6, 3)$, $K_{3,3}$ fails to be a maximiser exactly for $\gamma \in [1/18, 1/9)$	§1
SmallestCounterexample	$(6, 3, 1/18)$ is the smallest counterexample	§1, R1
Remark9	Remark 9	§6
Remark10a	Remark 10: for $\gamma > 1/8$, $i_\gamma(mK_{d,d}) = (1 - o(1))2^n$, and the relative difference is at most $U(mK_{d,d})/i_\gamma(mK_{d,d})$, which is at most Cc^m with $c < 1$	§7
Remark10b	Remark 10: the value $1.3 \cdot 10^{-17}$	§7
Remark10c	Remark 10: $q^* > 1$ exists, the base tends to 1 as $\gamma \downarrow 1/8$, and the values 1.0006 and 1.0001	§7
Remark10d	Remark 10: the mean $dn/8$ and the variance $dn(2d+1)/32$	§7
ReadingR6	reading R6 holds for $\gamma = 0$ and for $\gamma > 1/8$	§2
Remark11	Remark 11	§7
Remark12	Remark 12	§7
RegularOn2dConnected	every d -regular graph on $2d$ vertices is connected	§8
HighThresholdTie	for $d \geq 3$ all d -regular graphs on n vertices have the same $N_{\leq t}$ when $t \geq dn/2 - 2d$	§8
TwoSwitchConnected	two graphs with the same degree at every vertex are joined by a sequence of 2-switches [7, 6]	§8
Table2	Table 3 and the text after it: the numbers 2, 6, 60 of classes, the maxima, and the thresholds with a unique maximiser or a tie	§8
LabelledCounts	there are 70 labelled cubic graphs on six vertices	§8
TheoremDNumerics	the comparisons for Theorem D in Section 8	§8

How the proofs work. Most proofs follow the written ones. The edge counts of disjoint unions, of copies and of $K_{d,d}$ come from the identity $2e_G(A) = |\{(u, v) : u, v \in A, u \sim v\}|$, which also gives $P_{G \cup F} = P_G P_F$, $P_{K_{d,d}} = P_d$ and $P_{mK_{d,d}} = P_d^m$. Proposition 2 and Lemma 3 are proved as in Section 4: the 3-sets are counted through the cherries at their apex, and the heavy 4- and 5-sets through the cliques they contain. For $d = 2$, a 4-set spanning four edges is shown to be the vertex set of a component; this gives $Q(G) = c_4(G)$ and $4Q(G) \leq n$, and when $4Q(G) = n$ it gives an isomorphism $G \cong \frac{n}{4}K_{2,2}$ assembled from the components. The same lemma on blocks gives the trivial case $d = 1$. For Theorem B and Corollary 5, the formula for $e_{S_d}(A)$ of Section 5 is proved for every choice of x_1, x_2, y_1, y_2 , and the four families of extra sets and the independent sets of S_d are counted directly; the list of pairs (d, m) is checked for $d \leq 13$ and follows from $2^{d+1} > 2d^2 + 8$ for $d \geq 14$. The isomorphisms $S_3 \cong K_3 \square K_2$ and $H_2 \cong C_8$ are explicit for one choice of labels, and are transported to every other choice by automorphisms of $K_{d,d}$ and of $2K_{d,d}$. The 2-switches are handled by an edge-count identity valid for any 2-switch. Lemma 6 is proved from the identity $z^{e_{2K}(A)} - z^{e_{H_d}(A)} = (z-1)(\alpha_1 - \alpha_2)(\beta_1 - \beta_2)z^{e'(A)}$ for every set A , where $e'(A)$ counts the edges other than $u_1w_1, u_2w_2, u_1w_2, u_2w_1$, summed as in Section 5, and Lemma 7 from $4D(q) = \Delta$. The mean and variance in Remark 10 are computed with the signs $z_v(A) = \pm 1$, where $[v \in A] = (1 + z_v(A))/2$: a product of signs sums to zero over all sets A unless every vertex occurs in it an even number of times.

Several cited results are proved rather than assumed: the case of the theorem of Sah,

Sawhney, Stoner and Zhao that we use, and with it the theorem of Kahn and Zhao (its case $q = 0$) and Carenini’s Proposition 2.1 with fugacity 1; the 2-switch theorem; and the lower bound in Cramér’s theorem, Lemma 8. The bound $P_G(q) \leq P_d(q)^{n/(2d)}$ is proved for $0 \leq q \leq 1$, which at $q = 0$ is the theorem of Kahn and Zhao, in two steps. First, $P_G(q)^2 \leq Z(q)$, where $Z(q)$ is the sum of $q^{e(A,B)}$ over all pairs of vertex sets and $e(A, B)$ is the number of ordered adjacent pairs (u, v) with $u \in A$ and $v \in B$ (the edge polynomial of the bipartite double cover of G). Grouping the pairs (A, B) by $A \cap B$ and $A \triangle B$ reduces this to the inequality $\sum_{X \subseteq M} q^{u(X)} \leq \sum_{X \subseteq M} q^{c(X)}$, where $u(X)$ and $c(X)$ count the edges inside a set M that the cut $(X, M \setminus X)$ leaves uncut and cuts; this follows by expanding both products over sets of edges, in the spirit of Zhao’s bipartite swapping trick [14]. Second, $Z(q) = \sum_A \prod_v (1 + q^{|A \cap N(v)|}) \leq P_d(q)^{n/d}$ by Finner’s form of Hölder’s inequality [5] on the Boolean cube, as each vertex lies in exactly d neighbourhoods. The 2-switch theorem [7, 6] is proved by making the neighbourhoods agree one vertex at a time. Lemma 8 is proved without Cramér’s theorem, by exponential tilting and Chebyshev’s inequality for the tilted law: in part (a), the intermediate value theorem gives a tilt λ for which the tilted mean is $x + \delta$, and then the coefficients of $(\sum_i p_i z^i)^m$ within $m\delta$ of $m(x + \delta)$ have total mass at least $e^{m(\Lambda(\lambda) - \lambda(x + \delta) - \delta|\lambda|)} (1 - M^2/(m\delta^2))$; the law of Σ_m is identified with these coefficients by induction on m . Proposition C, Theorem D and Remarks 9–11 follow Sections 6 and 7, with Lemma 8 applied to independent copies of ξ on a product probability space. In Proposition C, the central limit theorem is not used for $\gamma \geq 1/8$: Lemma 8(b) at points just below $\mathbb{E}\xi = d^2/4$, with the bounds $P_d(q) \geq 4^d q^{d^2/4}$ (Jensen’s inequality) and $P_d(q) \geq P_d(0)$, gives the lower bound $\log 2$. In Remark 10, the limit $i_\gamma(mK_{d,d})/2^n \rightarrow 1$ and its exponential rate come from a Markov bound at some $q > 1$; the limit of the base as $\gamma \downarrow 1/8$ comes from the equation $\varphi'(q^*) = 0$ and the monotonicity of the tilted mean $qP'_d(q)/P_d(q)$; and the two values for $d = 2$ come from the enclosures $9/5 < q^* < 181/100$ at $\gamma = 1/4$ and $53/5 < q^* < 1063/100$ at $\gamma = 49/100$, with $P_{H_2}(q) = P_2(q)^2 - 2(q - 1)^4$ from Lemma 6. The counts on graphs with at most twelve vertices (Tables 1 and 4) are computed by the kernel on bit masks: a graph on $\{0, \dots, n - 1\}$ is encoded by the mask of its edges, a checker computes $N_{\leq t}$ of a mask by a loop over all vertex masks, and a lemma proves that the checker computes $N_{\leq t}$ of the graph; the graphs of the paper are identified with their masks by explicit isomorphisms. The cubic graphs on six vertices are enumerated over all 2^{15} edge masks, and each of the 70 cubic ones is mapped onto $K_{3,3}$ or the prism by an explicit permutation, found by a program and checked in Lean. For Table 3, the 2-switch theorem reduces the classification to a finite check: for each of the 2, 6 and 60 representatives and each of its 2-switches, an explicit isomorphism onto a representative is checked by the kernel (30, 307 and 7597 certificates), so every d -regular graph on $2d$ vertices is isomorphic to a representative. The representatives are pairwise non-isomorphic because their profiles $(N_{\leq t})_t$ differ, except for seven pairs with $d = 5$, which are separated by two further isomorphism invariants; the maxima, the unique maximisers and the ties are read off the profiles, which the kernel computes in a similar way. The comparisons of Remarks 10 and 12 and of Section 8 are exact computations with polynomials: i_γ of $mK_{d,d}$ and of G_n is a sum of initial coefficients of P_d^m or of $P_{H_d}^{\lfloor m/2 \rfloor} P_d^{m \bmod 2}$, where $P_{H_d} = P_d^2 - 2(z - 1)D$ by Lemma 6, and these powers are computed by the kernel as lists of coefficients, with multiplication and truncation proved correct. One verified loop is run once for each statement and each value of d (six runs), and each run checks the comparisons of its statement. Finite checks use `decide` and kernel reduction; there is no `native_decide`.

Checks. For every statement of Table 5, the recorded axiom audits (`#print axioms`) report only the axioms `propext`, `Classical.choice` and `Quot.sound`. A static scan found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern`, `unsafe`, and no axiom declaration, in the proof files. The Lean package of this paper, 94 modules (the statement file, its tests, the 91 proof modules and the check file), was built module by module from an empty directory, reusing only the pinned toolchain and the compiled Mathlib and its

dependencies, and every module was checked with `leanchecker`, all with exit code 0, twice (on 29 September 2026, between 03:19 and 04:16 UTC). For the release, the whole Lean project of the repository was rebuilt from a clean directory and all its modules were checked again with `leanchecker` (29 September 2026, 12:49–13:54 UTC). The statement file, its tests, the proofs, the check file and the two programs that generated the data of the finite checks (the data are checked by the kernel, not trusted) are in version 1.10.0 of the repository [4], directory `lean/Research/Backfill/Paper3/`; the receipt of the two builds is in `certificates/carenini/lean-package-v2/`, and the logs of the rebuild of the whole project are in `logs/`.

Definitions the reader must trust. Apart from Mathlib’s graphs, regularity, isomorphisms, Cartesian products, disjoint sums, cycles, complete and complete bipartite graphs, cliques, bipartiteness, connectedness, connected components, cardinalities, polynomials, logarithms, limits, infima, integrals, probability measures, independence and laws of random variables, the statements use the following definitions, each at most a few lines long: `edgesIn`, `iCount`, `NleZ`, `iGamma`, `boundary`, `edgePoly`, `triangles`, `Qcount` and `c4` (the quantities $e_G(A)$, $N_{\leq t}$ for natural and integer t , i_γ , $\partial(S)$, P_G , $T(G)$, $Q(G)$ and $c_4(G)$); `Kdd`, `copies`, `KddUnion`, `twoSwitch`, `Sd`, `Hd`, `GnV`, `Gn`, `prism`, `threeK4` and `prismUnion` (the graphs of the paper, where `copies m G` is the Cartesian product of the empty graph on m vertices with G , and a 2-switch deletes two edges and adds two with Mathlib’s `deleteEdges` and `fromEdgeSet`); `CareniniQuestion`, its bipartite form, `Question13`, `Admissible`, `tStar` and `gammaStar`; the polynomials `Pd`, `Wpoly` and `Dpoly` (P_d , $W_{\alpha\beta}$, D) and their real values (`evalR`), the functions `rhoFun` and `rho` ($\rho(d, \gamma)$), `phiFun`, `IsQStar` and `baseRatio` (the function φ , its minimisers q^* over $[1, \infty)$ and the base of Remark 10); and the predicates of Section 8: a 2-switch step, the number of isomorphism classes (a list of pairwise non-isomorphic representatives such that every graph with the property is isomorphic to one of them), failure, maximum, unique maximiser and tie at a threshold, a row of Table 3 (`Table2Row`), a win or a loss of G_n , and the list of the five values of γ at $n = 192$ (`gammasWin192`). A file of known-answer tests checks the counting and graph definitions, the polynomials, the 2-switch step and the thresholds on small cases from the paper, including instances that must come out false; `c4`, `rhoFun`, `rho`, `phiFun`, `IsQStar`, `baseRatio`, `prismUnion`, the predicates of the question and the other predicates of Section 8 have no separate test.

Faithfulness and non-claims. A graph is a Mathlib `SimpleGraph` on a finite type, and “a graph on n vertices” means that the type has n elements; so the statements cover graphs on every finite vertex type. “ $G \neq mC_4$ ” and “other than $K_{d,d}$ ” mean “not isomorphic to”. The graphs S_d , H_d and G_n are defined for every choice of the vertices x_1, x_2, y_1, y_2 and u_1, w_1, u_2, w_2 , and the statements hold for every choice, except the numerical statements `Table4Counts`, `Remark10b`, `Remark12` and `TheoremDNumerics` and the two values in `Remark10c`, which use one fixed choice (u_k and w_k the first vertices of their sides). A lower limit $\liminf_m \frac{1}{m} \log a_m \geq L$ is stated as: for every $\varepsilon > 0$, eventually $a_m \geq e^{m(L-\varepsilon)}$. Lemma 8 is stated for every family of independent random variables with the law of ξ , on any probability space. “Beats” and “loses” are strict inequalities. A value stated as “about” is read as a half-open interval of the last digit: about $1.3 \cdot 10^{-17}$ means in $[1.25, 1.35) \cdot 10^{-17}$, and about 1.0006 and 1.0001 mean in $[1.00055, 1.00065)$ and $[1.00005, 1.00015)$. The sentence on $n = 192$ in Section 8 is read as holding for each $d \in \{2, 3, 4\}$. `SSSZBound` and `CareniniProp21` are the special cases of [11, Corollary 1.15] and [2, Proposition 2.1] that the paper uses, not these results in general. The statement `LowGammaExamples` consists of two arithmetic facts; that they cover all our counterexamples with $\gamma \leq 1/8$ is the case analysis in Section 1 (Scope). The agreement of the Lean definitions with those of the paper and of [2] was checked by the reviewing agents, not by a machine.

Changes in version 2. Apart from this section, the disclosure and the dates, version 2 differs from version 1 as follows. Two claims for which we have no formal proof were deleted: the approximation $i_{1/8}(G) = (1/2 + O(n^{-1/4})) 2^n$ in Remark 10, which rested on a normal approximation bound, with the claim that reading R6 holds at $\gamma = 1/8$ (Table 2 and open question 3); and the count of 19,355 labelled 4-regular graphs on eight vertices in Section 8. Three remarks that were not formalised were removed: that Remark 4 is also the equality case of Mantel’s theorem; that an explicit n_0 in Theorem D could be extracted from quantitative forms of Cramér’s theorem; and, in open question 1, a remark on where the infimum defining $\rho(d, \gamma)$ is attained. Sentences saying that our counterexamples are consistent with other statements were replaced by the statements themselves (abstract, Section 1, Remarks 10 and 11, Section 8); Section 1 now gives the case analysis behind the claim that our counterexamples with $\gamma \leq 1/8$ have $\gamma d < 1$; reading R1 now says that the minimality of $(6, 3, 1/18)$ is also formally verified; and Section 8 says which numbers come from programs of version 1. Section 1 also reports the searches for version 2, and a few sentences were reworded for precision or, like “Our counterexamples are exact-count effects”, deleted as vague. The theorems and their written proofs are unchanged.

10 Open questions

1. (Reading R5.) Let $d \geq 2$ and $0 < \gamma \leq 1/8$ be fixed. Is $i_\gamma(G) \leq i_\gamma(mK_{d,d})$ for every d -regular graph G on n vertices, for all sufficiently large n divisible by $2d$? At $\gamma = 1/10$ the inequality fails for many n up to 400 (Remark 12), but the competitors used there lose for large n (Remark 11), and Proposition C shows that the answer is positive at the exponential scale. We make no claim.
2. All our counterexamples with $\gamma \leq 1/8$ have $\gamma d < 1$. Is there a counterexample, for some n , with $\gamma \leq 1/8$ and $\gamma d \geq 2$, the range of small γ in Carenini’s Proposition 1.2?
3. (Reading R6.) For fixed d and $0 < \gamma < 1/8$, is the maximum of i_γ over d -regular graphs on n vertices at most $(1 + o(1)) i_\gamma(mK_{d,d})$ as $n \rightarrow \infty$? (We make no claim at $\gamma = 1/8$.)
4. Which d -regular graphs maximise i_γ when $1/8 < \gamma < 1/2$ and n is large? By Proposition 2, at $\gamma = \gamma^*$ the maximisers are the graphs that maximise $2T(G) - Q(G)$.

Tool and computational resource disclosure

Version 1 of this paper was produced on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. The author set the research programme, namely resolving open problems with AI agents and formal verification, and directed the agents; the agents selected this question. Claude Code agents found the counterexamples and the proofs, wrote the verification programs and the Lean formalisation of version 1, carried out the literature searches described in the introduction, and drafted the text. An independent Claude Code agent, without access to the reasoning of the agents that found the proofs, refereed the proofs and the correspondence between the Lean statements and the question, and recomputed the main numbers with its own programs. A second independent Claude Code agent then reviewed the whole paper: it checked the proofs and the statements against the Lean file, recomputed the numbers with its own programs, and checked the references. Before release, OpenAI Codex (desktop application) carried out a brief independent check of the question, the main statements and their Lean formalisation, and recounted the numbers 28 and 24; its one finding, that Theorem A must require $n > 0$, was applied.

Version 2 was produced from 27 to 29 September 2026. A Claude Code agent (model Claude Opus 5.5) wrote version 1 of the statement file of Section 9, and an independent Claude Code agent reviewed it and recomputed the numbers of the paper with its own programs. OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`) checked version 1 of the paper independently, recomputing most of its numbers with its own programs, and audited version 1 of the statement file. Codex’s usage quota then ran out, and Claude Code carried out the rest: it wrote version 2 of the statement file, which an independent Claude Code agent reviewed; on 29 September 2026 Claude Code agents wrote the Lean proofs, working in parallel, and Claude Code ran the checks described in Section 9. An independent Claude Code agent reviewed the Lean package and checked the account of it in Section 9 against the code. Claude Code drafted the changes of version 2, and another independent Claude Code agent refereed a draft of version 2 and recomputed the main numbers with its own programs. Codex has not reviewed version 2 of the paper, version 2 of the statement file or the Lean proofs. Throughout, “we” refers to this workflow. The author is not a professional mathematician. The author has read version 2 of the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 5 and on the agreement of those statements with the paper, which was checked by the reviewing agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

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